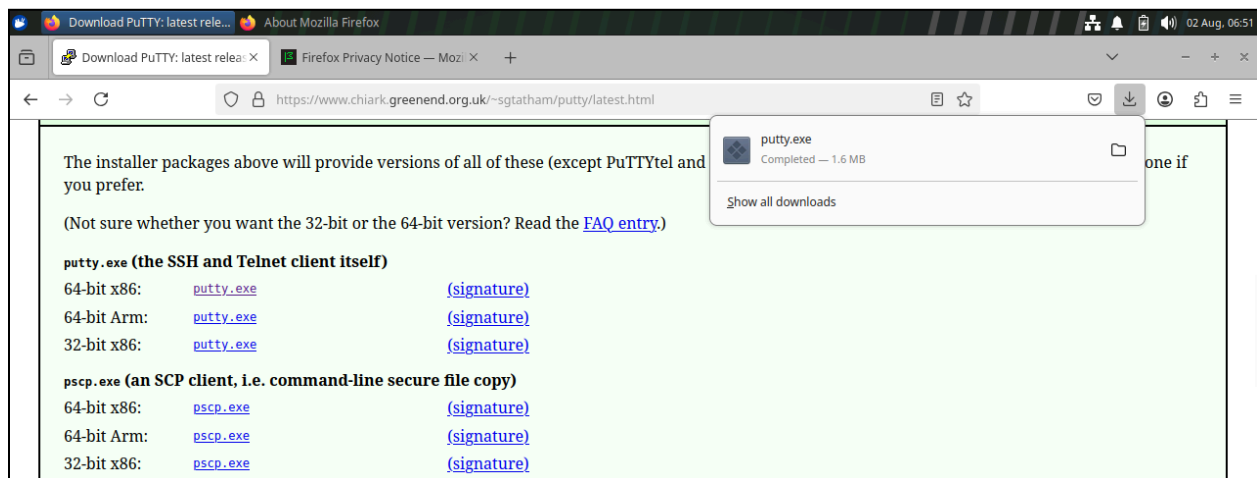


Malware Analysis A2

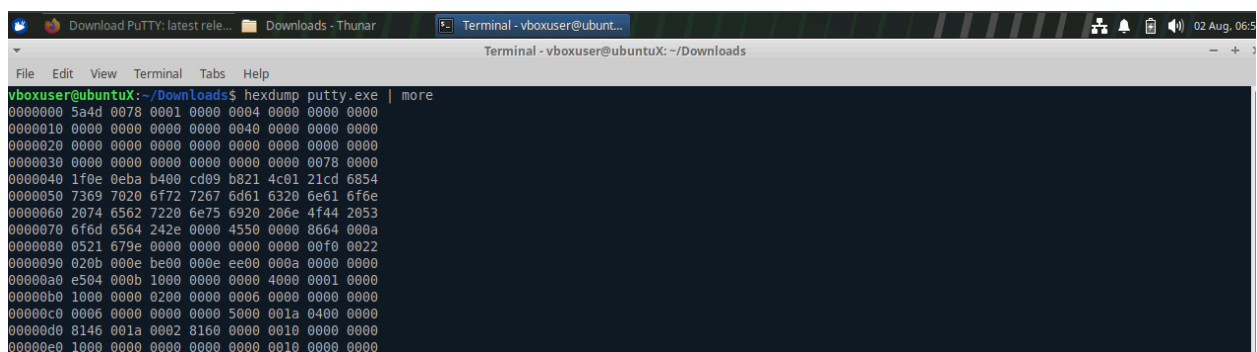
Objective:

Download a sample .exe (PE) file and an ELF binary file (not malware). Use hexdump to view the raw binary layout, and run strings to extract human-readable content. Check if any URLs, messages, or other readable data are present in both files.

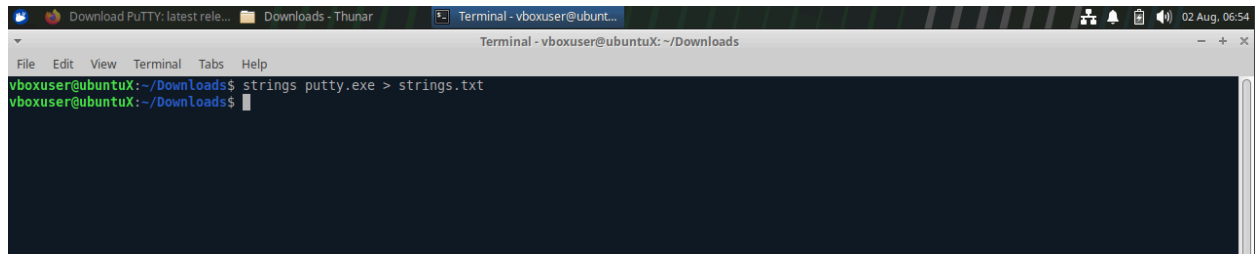
A sample file was downloaded. The file acquired was **putty.exe**, the SSH and Telnet client, from the website www.greenend.org.uk



hexdump putty.exe | more was executed to inspect the raw binary layout of the file.

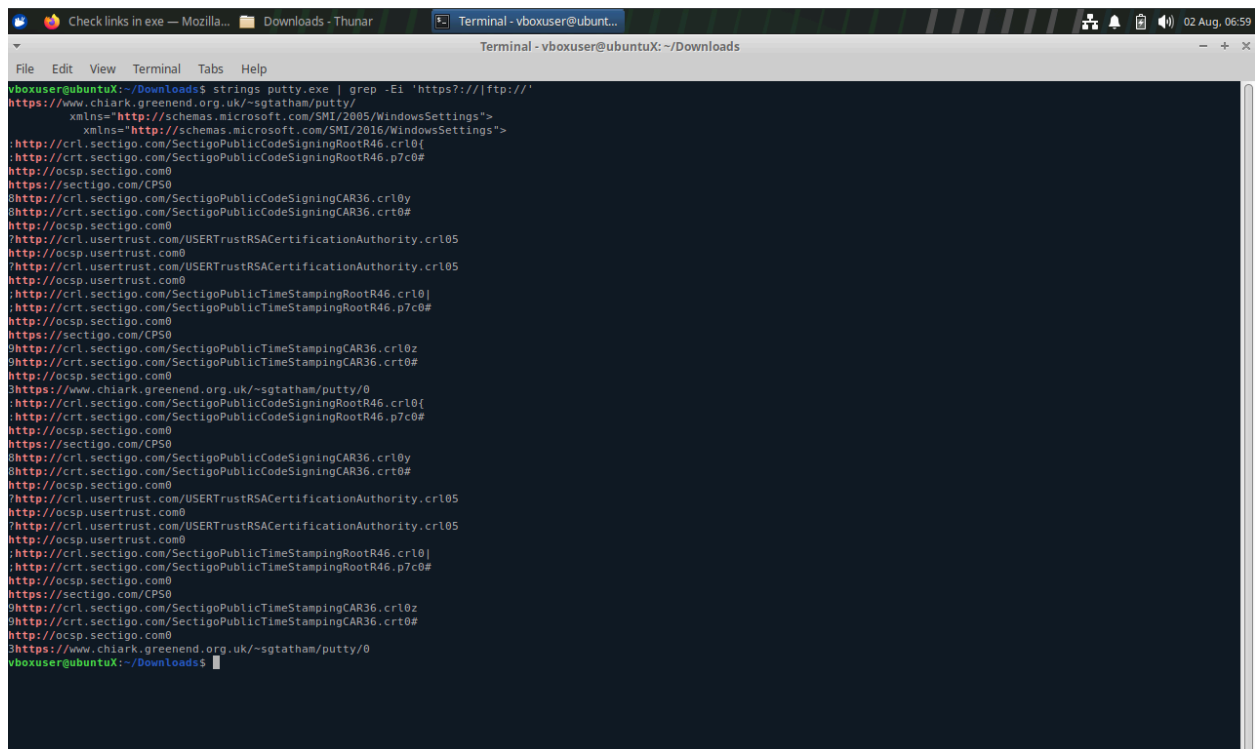


strings putty.exe > strings.txt was run, which took all the printable strings from the executable and stored the result in a file named strings.txt.



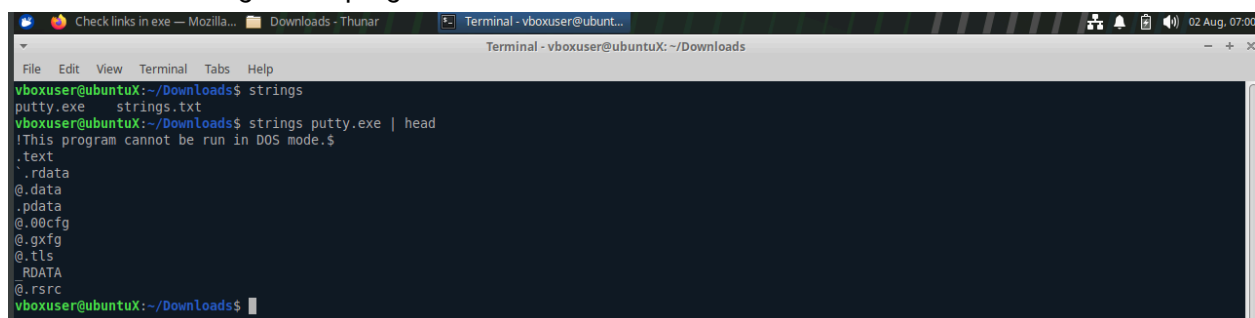
```
Terminal - vboxuser@ubuntuX: ~/Downloads
File Edit View Terminal Tabs Help
vboxuser@ubuntuX:~/Downloads$ strings putty.exe > strings.txt
vboxuser@ubuntuX:~/Downloads$
```

Once the strings were extracted, the next step was to check if any URLs or data were available in the file. A search of the strings revealed multiple URLs. These included the official PuTTY project URL



```
Terminal - vboxuser@ubuntuX: ~/Downloads
File Edit View Terminal Tabs Help
vboxuser@ubuntuX:~/Downloads$ strings putty.exe | grep -Ei 'https?://|ftp://'
https://www.chiark.greenend.org.uk/~sgtatham/putty/
  xmlns="http://schemas.microsoft.com/SMI/2005/WindowsSettings">
  xmlns="http://schemas.microsoft.com/SMI/2016/WindowsSettings">
;http://crl.sectigo.com/SectigoPublicCodeSigningRootR46.crl0{
;http://crl.sectigo.com/SectigoPublicCodeSigningRootR46.p7c0#
http://ocsp.sectigo.com0
https://sectigo.com/CP50
8http://crl.sectigo.com/SectigoPublicCodeSigningCAR36.crl0y
8http://crl.sectigo.com/SectigoPublicCodeSigningCAR36.crt0#
http://ocsp.sectigo.com0
?http://crl.usertrust.com/USERTrustRSACertificationAuthority.crl05
http://ocsp.usertrust.com0
?http://crl.usertrust.com/USERTrustRSACertificationAuthority.crl05
http://ocsp.usertrust.com0
;http://crl.sectigo.com/SectigoPublicTimeStampingRootR46.crl0{
;http://crl.sectigo.com/SectigoPublicTimeStampingRootR46.p7c0#
http://ocsp.sectigo.com0
https://sectigo.com/CP50
8http://crl.sectigo.com/SectigoPublicTimeStampingCAR36.crl0z
8http://crl.sectigo.com/SectigoPublicTimeStampingCAR36.crt0#
http://ocsp.sectigo.com0
https://sectigo.com/CP50
8https://www.chiark.greenend.org.uk/~sgtatham/putty/0
;http://crl.sectigo.com/SectigoPublicCodeSigningRootR46.crl0{
;http://crl.sectigo.com/SectigoPublicCodeSigningRootR46.p7c0#
http://ocsp.sectigo.com0
https://sectigo.com/CP50
8http://crl.sectigo.com/SectigoPublicCodeSigningCAR36.crl0y
8http://crl.sectigo.com/SectigoPublicCodeSigningCAR36.crt0#
http://ocsp.sectigo.com0
?http://crl.usertrust.com/USERTrustRSACertificationAuthority.crl05
http://ocsp.usertrust.com0
?http://crl.usertrust.com/USERTrustRSACertificationAuthority.crl05
http://ocsp.usertrust.com0
;http://crl.sectigo.com/SectigoPublicTimeStampingRootR46.crl0{
;http://crl.sectigo.com/SectigoPublicTimeStampingRootR46.p7c0#
http://ocsp.sectigo.com0
https://sectigo.com/CP50
8http://crl.sectigo.com/SectigoPublicTimeStampingCAR36.crl0z
8http://crl.sectigo.com/SectigoPublicTimeStampingCAR36.crt0#
http://ocsp.sectigo.com0
8https://www.chiark.greenend.org.uk/~sgtatham/putty/0
vboxuser@ubuntuX:~/Downloads$
```

The **head** command was used to retrieve the initial data from the strings. **strings putty.exe | head** showed the message "This program cannot be run in DOS mode."



```
Terminal - vboxuser@ubuntuX: ~/Downloads
File Edit View Terminal Tabs Help
vboxuser@ubuntuX:~/Downloads$ strings
putty.exe
vboxuser@ubuntuX:~/Downloads$ strings putty.exe | head
!This program cannot be run in DOS mode.$
.text
.rdata
@.data
.pdata
@.00cfg
@.gxf
@.tls
RDATA
@.rsrc
vboxuser@ubuntuX:~/Downloads$
```